Voltage: energy per charge

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Interpret potential difference as energy transferred per unit charge.

SUCCESS CRITERIA - I can...

1. Use volts as joules per coulomb.
2. Calculate energy transfer with E=qV.
3. Distinguish voltage from current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “voltage” and “charge” to compare your model or result. Sentence starters are optional.

voltage: electric potential difference; energy transferred per charge

charge: a property responsible for electric interactions

energy: a quantity tracked through storage and transfers in a system

01 Warm-up: bring back an earlier idea

A neutral object is attracted to an object carrying +3 nC. **Explain** how attraction can occur without giving the neutral object a net charge.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: energy transferred = qV; V = energy transferred/q. Magnitudes here; 1 V = 1 J/C. Voltage is a difference between two points.

Worked reasoning: Moving 2 C through a potential difference of 3 V changes electric potential energy by q ΔV=6 J. Voltage is energy change per charge, not energy by itself.

Your turn: Draw two labeled potential levels and an energy account for a positive charge moved between them.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial / supply _____ V: ____ J/C

Final / component _____ V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Compare a 3 V and a 6 V battery for the same amount of charge transferred. Which supplies more energy per coulomb?

Sentence starter: Both … . They differ in … because … .

Initial / supply

_____

V: ____ J/C

Final / component

_____

V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: 1 2 3

04 Apply the physics

A battery supplies 240 J while 20 C passes through it.
Find its potential difference. In a separate component, 5 C passes through a 3 V drop.
Calculate the energy transferred there and label the two cases.

Sentence starter: Both … . They differ in … because … .

Initial / supply

_____

V: ____ J/C

Final / component

_____

V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student says a 9 V battery contains 9 coulombs.
Explain the units and why voltage does not state total battery capacity.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Sensors 1.36 m apart record delay 0.004 s.
Find wave speed. If 0.001 s is added to the delay, find the incorrect speed and describe the bias.
Label the full travel path and the recorded time.

Sentence starter: The step I would change is … . The evidence is … .



Marzano check - circle: 1 2 3

Voltage: energy per charge

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Two components in series have drops 2 V and 4 V. For 3 C passing through each, find each energy transfer and the total.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial / supply _____ Final / component _____

V: ____ J/C V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A charge -2 microcoulombs moves through ΔV=+50 V.
Find ΔU and work by the electric field.
Explain both signs.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Initial / supply _____ Final / component _____

V: ____ J/C V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: 1 2 3